Therapeutic AI Chatbot - Project Plan

🎯 Vision

Create an accessible therapeutic bot that provides clinical-grade therapeutic support for individuals with mental health challenges (depression, anxiety, PTSD, etc.). The system intelligently transitions users to real clinical therapists when their needs exceed AI capabilities.

📊 Project Phases

Phase 0: Foundation (Current) ☑

Duration: 1-2 weeks | **Status:** In Progress

✓ Completed

- Safety checks for crisis keywords
- Multi-mode therapeutic responses (Witness, Companion, Gentle Guide, Quiet Presence)
- FastAPI backend structure
- Basic schema definitions (ChatRequest, ChatResponse)

🔨 Immediate Tasks

- ☐ Add persistent session storage (PostgreSQL/SQLite)
- ☐ Create user authentication system
- ☐ Add logging and monitoring
- ☐ Create frontend (React/Vue) or mobile app interface
- ☐ Deploy to staging environment

Tech Stack

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Backend: FastAPI, Pydantic
Database: PostgreSQL/SQLite
Frontend: React.js or React Native
Deployment: Docker, AWS/GCP
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Phase 1: Enhanced Therapeutic AI (4-6 weeks)

Core Features

1.1 LLM Integration

- ☐ Integrate OpenAI API (GPT-4) or Anthropic Claude
- ☐ Create custom system prompts for each mode with clinical grounding

- ☐ Add response streaming for better UX
- ☐ Implement token management and cost optimization

1.2 Session Management

- ☐ Build session persistence layer
- ☐ Store conversation history with timestamps
- ☐ Implement user profile system tracking:
  - Primary mental health concerns
  - Therapy mode preferences
  - Support network information
  - Crisis contact details

1.3 Escalation Detection

- ☐ Expand crisis keyword detection
- ☐ Implement symptom severity scoring
- ☐ Create escalation threshold logic
- ☐ Build escalation notification system
- ☐ Alert mechanism for support contacts

1.4 Therapeutic Technique Selection

- ☐ Implement CBT (Cognitive Behavioral Therapy) modules
- ☐ Add DBT (Dialectical Behavior Therapy) techniques
- ☐ Incorporate mindfulness/grounding exercises
- ☐ Create psychoeducational content delivery

Success Metrics

- Session completion rate > 80%
- User satisfaction score > 4/5
- Crisis detection accuracy > 95%
- Response generation < 2 seconds

Phase 2: Professional Network & Escalation (6-8 weeks)

Core Features

2.1 Medical Database Integration

- ☐ Connect to therapist/counselor databases:
  - PubMed/PsycINFO for evidence lookup
  - SAMHSA National Helpline directory
  - State licensing board databases
  - Insurance provider networks
- ☐ Build local professional directory crawler

2.2 Geolocation & Matching

- ☐ Implement location services integration
- ☐ Build therapist locator system with filters:
  - Specialization (depression, anxiety, trauma, etc.)
  - Insurance accepted
  - Availability & wait times
  - Client reviews/ratings
  - Language preferences
  - Virtual vs. in-person
- ☐ Create matching algorithm based on user needs

2.3 Escalation Workflow

- ☐ Build escalation decision tree
- ☐ Create professional recommendation UI
- ☐ Implement warm handoff system (alerts to therapists)
- ☐ Add crisis resource hotlines (988 Suicide & Crisis Lifeline, etc.)

2.4 Administrative Backend

- ☐ Therapist onboarding system
- ☐ Credential verification
- ☐ Performance tracking
- ☐ Feedback collection

APIs Required

- Google Maps / Mapbox for geolocation
- Third-party therapist databases
- Insurance verification APIs

Phase 3: Appointment Booking & Automation (8-10 weeks)

Core Features

3.1 Booking Integration

- ☐ Integrate with major therapy platforms:
  - Calendly
  - Therapy Practice Management Software (SimplePractice, TherapyLog)
  - Google Calendar API
- ☐ Build in-app booking interface
- ☐ Implement payment processing (Stripe)
- ☐ Handle insurance verification & billing

3.2 Communication System

- ☐ Phone call system (Twilio API):
  - Bot-to-user appointment confirmation
  - Bot-assisted therapist connection
- ☐ SMS reminders and follow-ups
- ☐ Email notifications
- ☐ In-app messaging

3.3 Appointment Management

- ☐ Reminder system (24h, 1h before)
- ☐ Rescheduling/cancellation workflow
- ☐ No-show tracking and penalties
- ☐ Follow-up session recommendations

3.4 Feedback & Continuous Improvement

- ☐ Post-session user surveys
- ☐ Therapist feedback on bot recommendations
- ☐ Outcome tracking (DASS-21, PHQ-9 scores)
- ☐ A/B testing framework for prompts

🏗 Architecture Overview

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┌──────────────────────────────────────────────────────────┐
│              Mobile/Web Frontend                         │
│            (React Native/React.js)                       │
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                   │
┌──────────────────▼───────────────────────────────────────┐
│              API Gateway                                 │
│            (FastAPI, Rate Limiting)                      │
└──────────────────┬───────────────────────────────────────┘
                   │
       ┌───────────┴──────────┐
       │          │          │
┌──────▼──┐  ┌────▼────┐  ┌──▼──────┐
│ Chat    │  │ Session │  │ User    │
│Engine   │  │Manager  │  │Service  │
│(LLM)    │  │         │  │         │
└─────────┘  └─────────┘  └─────────┘
       │          │          │
┌──────▼──────────▼──────────▼──────┐
│      PostgreSQL Database          │
│  (Sessions, Users, History)       │
└────────────────────────────────────┘
       │
┌──────▼──────────────────────────┐
│  External Services              │
│  ├─ OpenAI API (LLM)            │
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│  ├─ Therapist Directory API   │
│  ├─ Twilio (SMS/Phone)        │
│  ├─ Google Maps (Location)    │
│  └─ Payment Gateway (Stripe)  │
└──────────────────────────────┘
```

📋 Implementation Roadmap

Week 1-2 (Phase 0 Completion)

- ☐ Database schema finalization
- ☐ User authentication
- ☐ Session persistence
- ☐ Frontend scaffolding

Week 3-8 (Phase 1)

- ☐ LLM API integration
- ☐ Conversation history
- ☐ Enhanced escalation detection
- ☐ Therapy technique modules

Week 9-16 (Phase 2)

- ☐ Therapist directory integration
- ☐ Geolocation services
- ☐ Matching algorithm
- ☐ Recommendation engine

Week 17-26 (Phase 3)

- ☐ Booking system
- ☐ Phone/SMS integration
- ☐ Appointment management
- ☐ Analytics & reporting

🔒 Safety & Compliance

Critical Requirements

- ☐ HIPAA compliance (if handling PHI)
- ☐ GDPR compliance (EU users)
- ☐ CCPA compliance (California users)
- ☐ 24/7 crisis hotline integration
- ☐ Encrypted storage for sensitive data
- ☐ Audit logging for all interactions
- ☐ Mental Health Professional oversight

- ☐ Informed consent flows
- ☐ Data backup & disaster recovery

Risk Mitigation

- Always provide hotline numbers alongside AI responses
- Never provide diagnosis (only supportive listening)
- Escalate immediately for suicide/self-harm mentions
- Regular security audits
- A/B testing with mental health professionals

💼 Resource Requirements

Team

- 2-3 Backend Engineers
- 1-2 Frontend Engineers
- 1 Product Manager
- 1 Mental Health Professional Consultant
- 1 DevOps/Infrastructure Engineer

Infrastructure

- Cloud Platform (AWS/GCP/Azure)
- Database (PostgreSQL)
- Monitoring (DataDog/Sentry)
- CI/CD Pipeline (GitHub Actions)

Budget Considerations

- LLM API costs (~$0.01-0.05 per message)
- Cloud hosting (~$500-2000/month depending on scale)
- Third-party APIs and integrations
- Legal/compliance consulting
- Staff and contractor costs

📊 Success Metrics

Phase 1

- AI response quality score > 4.2/5
- Session completion rate > 75%
- Average session duration > 5 minutes

Phase 2

- Successful therapist matches > 80%
- Avg time to professional recommendation < 15 min

- Therapist acceptance rate > 60%

Phase 3

- Appointment booking success > 85%
- No-show rate < 20%
- User retention after first appointment > 70%

🎓 Learning Resources

- Mental Health First Aid Training
- CBT/DBT Framework Documentation
- HIPAA Compliance Training
- LLM Prompt Engineering Best Practices
- Crisis Intervention Protocols

📝 Notes

- Start with MVP focusing on chat + crisis detection
- Get mental health professional review before Phase 1
- Consider open-sourcing core algorithms
- Build community of therapist partners early
- Iterate based on user feedback and outcomes